General formulae

$$\operatorname{atan2}(x,y) = \begin{cases} \arctan \frac{y}{x} & \text{if } x > 0 \\ \arctan \frac{y}{x} + \pi & \text{if } x < 0 \text{ \& } y \geq 0 \\ \arctan \frac{y}{x} - \pi & \text{if } x < 0 \text{ \& } y < 0 \\ \frac{\pi}{2} & \text{if } x = 0 \text{ \& } y > 0 \\ -\frac{\pi}{2} & \text{if } x = 0 \text{ \& } y < 0 \end{cases}$$

$\cos(-\theta) = \cos\theta$; $\sin(-\theta) = -\sin(\theta)$

$\cos\left(x + \frac{\pi}{2}\right) = -\sin(x)$; $\cos\left(x - \frac{\pi}{2}\right) = \sin(x)$

$\sin\left(x + \frac{\pi}{2}\right) = \cos(x)$; $\sin\left(x - \frac{\pi}{2}\right) = -\cos(x)$

	0	30	45	60	90	180	270	120
sin	0	$1/2$	$\sqrt{2}/2$	$\sqrt{3}/2$	1	0	-1	$\frac{\sqrt{3}}{2}$
cos	1	$\frac{\sqrt{3}}{2}$	$\sqrt{2}/2$	$1/2$	0	-1	0	$-\frac{1}{2}$
tan	0	$1/\sqrt{3}$	1	$\sqrt{3}$	∞	0	undef.	$-\sqrt{3}$

<u>Derivatives</u>

$c_1 c_2 = -\dot{q}_1 s_1 c_2 - \dot{q}_2 c_1 s_2$ $\qquad$ $s_1 c_2 = \dot{q}_1 c_1 c_2 - \dot{q}_2 s_1 s_2$

$s_1 s_2 = \dot{q}_1 c_1 s_2 + \dot{q}_2 s_1 c_2$ $\qquad$ $c_1 s_2 = -\dot{q}_1 s_1 s_2 + \dot{q}_2 c_1 c_2$

→ dont write $\dot{q}_1, \dot{q}_2$ for partial derivatives

$\frac{\partial}{\partial q_1} c_{12} = \frac{d}{dq_1}(\cos(q_1 + q_2)) = -\sin(q_1 + q_2) \cdot 1$

$\frac{\partial}{\partial q_1} s_{12} = \frac{d}{dq}(\sin(q_1 + q_2)) = \cos(q_1 + q_2) \cdot 1$

$(uv)' = u'v + uv'$

$(uvw)' = u'vw + uv'w + uvw'$

Assignment of COS in DH convention

1. axes X_1 & Z_0 are orthogonal

2. " X_1 & Z_0 interset

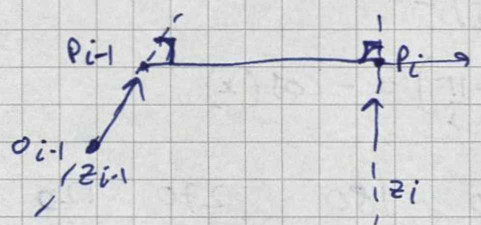
step 1: choose z_i to determine motion of joint $i+1$

step 2: choose base COS arbitrarily along Z_0

choose x & y with Right hand COS.

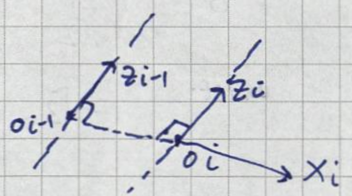
a) if 2 axes are not coplanar $[z_{i-1}$ & $z_i]$

i.e. 2 axes that never intersect



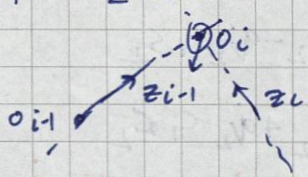
choose x_i in directin of shortest connecting line & place o_i at p_i

b) if axes are $\|^{le}$



choose o_i on z_i with min. dist from preceeding origin.

c) if 2 axes intersect



place o_i at point of intersection of 2 axes

choose x_i as cross product of z_{i-1} & z_i

Scalar product is commutative : $x^T y = y^T x$

$$(R_1^0)^{-1} = (R_1^0)^T = R_0^1$$

Special orthogonal group: SO(n)
- all columns are mutually orthogonal
- $R^T = R^{-1}$
- det $R = 1$

$$R_1^0 = \begin{bmatrix} x_1^T x_0 & y_1^T x_0 & z_1^T x_0 \\ x_1^T y_0 & y_1^T y_0 & z_1^T y_0 \\ x_1^T z_0 & y_1^T z_0 & z_1^T z_0 \end{bmatrix}$$

$$R_{z,\theta} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad R_{x,\theta} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix} \quad R_{y,\theta} = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

Similarity transform :

$$B = (R_1^0)^{-1} A R_1^0$$

$A \rightarrow$ linear mapping in $o_0 x_0 y_0 z_0$

$B \rightarrow$ same linear mapping in $o_1 x_1 y_1 z_1$

Rotation WRT COS : multiply on right

$$p^0 = R_1^0 p^1 \qquad p^1 = R_2^1 p^2$$

$$p^0 = R_1^0 R_2^1 p^2$$

$$\Rightarrow \boxed{R_2^0 = R_1^0 R_2^1}$$

Rotation WRT fixed COS : multiply on left

$$\boxed{R_2^0 = R_1^0 \left((R_1^0)^{-1} R R_1^0\right) = R R_1^0}$$

↳ $R =$ r.. WRT fixed CO..

Homogeneous transformation: orintation + position = rigid motion

$$p^0 = R_1^0 p^1 + o_1^0 \qquad p^1 = R_2^1 p^2 + o_2^1$$

$$p^0 = \underbrace{R_1^0 R_2^1}_{\text{rot. matrix}} p^2 + \underbrace{R_1^0 o_2^1 + o_1^0}_{\text{translation vector}} = R_2^0 p^2 + o_2^0$$

$$\bar{p}^0 = \begin{bmatrix} R_1^0 & o_1^0 \\ 0^T & 1 \end{bmatrix} \bar{p}^1 = T_1^0 \bar{p}^1$$; $T \rightarrow$ homogeneous transform

$$\bar{p}^0 = T_1^0 \bar{p}^1 = T_1^0 T_2^1 \bar{p}^2 = T_2^0 \bar{p}^2$$

Special Euclidean group : SE(n)

$$T = \begin{bmatrix} R & o \\ 0^T & 1 \end{bmatrix}$$

$$T^{-1} = \begin{bmatrix} R & o \\ 0^T & 1 \end{bmatrix}^{-1} = \begin{bmatrix} R^T & -R^T o \\ 0^T & 1 \end{bmatrix}$$ $\rightarrow o \rightarrow$ vector

↳ inverse of homogeneous transform

Elementary rotation matrices

$$\text{Trans}_{x,d} = \begin{bmatrix} 1 & 0 & 0 & d \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad \text{Rot}_{x,\theta} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c\theta & -s\theta & 0 \\ 0 & s\theta & c\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{Trans}_{y,d} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & d \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad \text{Rot}_{y,\theta} = \begin{bmatrix} c\theta & 0 & s\theta & 0 \\ 0 & 1 & 0 & 0 \\ -s\theta & 0 & c\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{Trans}_{z,d} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad \text{Rot}_{z,\theta} = \begin{bmatrix} c\theta & -s\theta & 0 & 0 \\ s\theta & c\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Euler angles [3 independent angles (ϕ, θ, ψ)] : each angle is rotation around x, y or z axis. Convention needs to be fixed.

ZYZ convention: $R_{ZYZ} = R_{z,\phi} R_{y,\theta} R_{z\psi}$

$$R_{ZYZ} = \begin{bmatrix} c_\phi c_\theta c_\psi - s_\phi s_\psi & -c_\phi c_\theta s_\psi - s_\phi c_\psi & c_\phi s_\theta \\ s_\phi c_\theta c_\psi + c_\phi s_\psi & -s_\phi c_\theta s_\psi + c_\phi c_\psi & s_\phi s_\theta \\ -s_\theta c_\psi & s_\theta s_\psi & c_\theta \end{bmatrix}$$

Roll, pitch & yaw [3 elementary rotations (ϕ, θ, ψ) around x_0, y_0 & z_0 relative to Fixed COS]

$$R = R_{z\phi} R_{y,\theta} R_{x,\psi} = \begin{bmatrix} c_\phi c_\theta & -s_\phi c_\psi + c_\phi s_\theta s_\psi & s_\phi s_\psi + c_\phi s_\theta c_\psi \\ s_\phi c_\theta & c_\phi c_\psi + s_\phi s_\theta s_\psi & -c_\phi s_\psi + s_\phi s_\theta c_\psi \\ -s_\theta & c_\theta s_\psi & c_\theta c_\psi \end{bmatrix}$$

Axis & Angle [4 parameters $k = [k_x, k_y, k_z]^T$ & angle θ]
↳ unit vector

$R_{k,\theta} = R_{z,\alpha} R_{y,\beta} R_{z,\theta} R_{y,-\beta} R_{z,-\alpha}$ → expansion on pg 26 [Eqn. 2.53]

$$\theta = \arccos\left(\frac{r_{11} + r_{22} + r_{33} - 1}{2}\right) ; \quad k = \frac{1}{2\sin\theta} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix}$$

Quaternions [4 parameters $k = [k_x \ k_y \ k_z]^T$ & angle θ]
↳ unit vector

$$Q = \left(\cos\frac{\theta}{2} \quad k_x \sin\frac{\theta}{2} \quad k_y \sin\frac{\theta}{2} \quad k_z \sin\frac{\theta}{2}\right) = \left(\cos\frac{\theta}{2} \quad k \sin\frac{\theta}{2}\right)$$

↳ unit quaternion expresses rotation by angle θ around unit vector $k = [k_x \ k_y \ k_z]^T$

$$Q = (q_0, q_1, q_2, q_3) ; \quad q_0 = \frac{1}{2}\sqrt{r_{11} + r_{22} + r_{33} + 1}$$

$$q_1 = \frac{r_{32} - r_{23}}{4 q_0} ; \quad q_2 = \frac{r_{13} - r_{31}}{4 q_0} ; \quad q_3 = \frac{r_{21} - r_{12}}{4 q_0}$$

Multiplication of 2 quaternions

$$X = (x_0, x_1, x_2, x_3) \quad Y = (y_0, y_1, y_2, y_3) \quad Z = XY ; \quad XY \neq YX$$

$$z_0 = x_0 y_0 - x^T y ; \quad z = x_0 y + y_0 x + x \times y$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \times \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} x_2 y_3 - x_3 y_2 \\ x_3 y_1 - x_1 y_3 \\ x_1 y_2 - x_2 y_1 \end{bmatrix}$$

FWD kinematics: robots with n joints & $n+1$ links

each joint described by joint variable q_i.

q_i = translation → prismatic joint
rotation → revolute joint

$$q = \begin{bmatrix} q_1 \\ \vdots \\ q_n \end{bmatrix}$$ → description of complete kinematic chain

inertial system / base COS → $o_0 x_0 y_0 z_0$

$A_i(q_i)$ → homogeneous transformation from $o_i x_i y_i z_i$ relative to $o_{i-1} x_{i-1} y_{i-1} z_{i-1}$

↳ written also as A_i

$$T^i_j = \begin{cases} A_{i+1} A_{i+2} \cdots A_{j-1} A_j \ ; & \text{if } i<j \\ I \ ; & \text{if } i=j \\ \left(T^j_i\right)^{-1} \ ; & \text{if } i>j \end{cases}$$

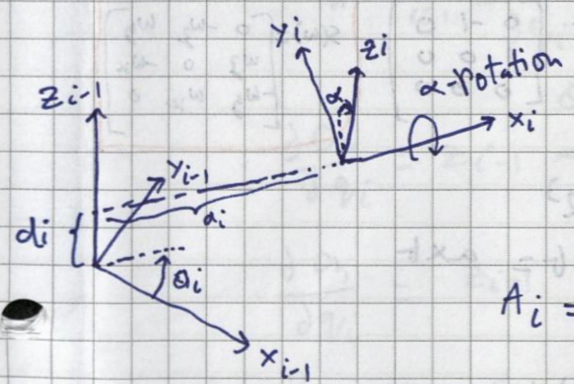
→ Transformation of arbitrary cos of link j WRT another link i.

$T^0_1 = A_1$
$T^1_2 = A_2$

$$T^0_n = A_1 A_2 \cdots A_n = \begin{bmatrix} R^0_n & o^0_n \\ 0^T & 1 \end{bmatrix}$$ → pose of end effector

DH-convention: constraining placement of COS to reduce parameters.

$A_i = T^{i-1}_i$; i.e. from COS $o_i x_i y_i z_i$ to $o_{i-1} x_{i-1} y_{i-1} z_{i-1}$



a_i → link length
α_i → link twist
d_i → link offset
θ_i → joint angle

3 of 4 parameters/link are constant.

$$A_i = \text{Rot}_{z,\theta_i} \ \text{Trans}_{z,d_i} \ \text{Trans}_{x,a_i} \ \text{Rot}_{x,\alpha_i} \ ; \quad i = 1 \ldots n$$

$$= \begin{bmatrix} C_{\theta i} & -S_{\theta i} C_{\alpha i} & S_{\theta i} S_{\alpha i} & a_i C_{\theta i} \\ S_{\theta i} & C_{\theta i} C_{\alpha i} & -C_{\theta i} S_{\alpha i} & a_i S_{\theta i} \\ 0 & S_{\alpha i} & C_{\alpha i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- a is distance between z_0 & z_1 measured along x_1 axis
- α is angle between z_0 & z_1 measured in plane normal to x_1
- d is distance between origin and point of intersection of z_0 & x_1 axis measured along z_0 axis
- θ is angle between x_0 & x_1 measured in plane normal to z_0.

Assignment of cos for DH convention definition 2·6

1. choose z_i-axis that determines motion of joint $i+1$
2. define base cos $o_0 x_0 y_0 z_0$. Origin can arbitarily chosen along z_0 axis
3. X & Y axis chosen as per right hand rule.

Inverse kinematics

$\sin^2\theta + \cos^2\theta = 1$; $c_{12} = c_1 c_2 - s_1 s_2$; $s_{12} = c_1 s_2 + s_1 c_2$

$u = \tan\frac{q}{2}$; $\cos q = \frac{1-u^2}{1+u^2}$, $\sin q = \frac{2u}{1+u^2}$

$\cos(-\theta) = \cos(\theta)$; $\sin(-\theta) = -\sin(\theta)$; $\cos(x+\frac{\pi}{2}) = -\sin(x)$

$\cos(x-\frac{\pi}{2}) = \sin(x)$ $\sin(x+\frac{\pi}{2}) = \cos(x)$

$\sin(x-\frac{\pi}{2}) = -\cos(x)$

Decoupling decouples inverse pos. kin. & inverse orientation Kin.

Differential Kinematics

$S \rightarrow$ skew symmetric matrix

$$\boxed{S = \frac{dR(\theta)}{d\theta} R(\theta)^T} \quad ; \quad \frac{dR(\theta)}{d\theta} = S R(\theta)$$

$$S(a) = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix} \qquad S(w) = \begin{bmatrix} 0 & -w_z & w_y \\ w_z & 0 & -w_x \\ -w_y & w_x & 0 \end{bmatrix}$$

$$S(e_x) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \quad S(e_y) = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix} \quad S(e_z) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Properties of skew-symmetric Matrices

1. $\boxed{S(\alpha a + \beta b) = \alpha S(a) + \beta S(b)}$
   $\alpha, \beta \rightarrow$ scalars
   $a, b \in R^3 \rightarrow$ vectors
2. $a, b \in R^3$
   $S(a)\, b = a \times b$
3. for any $R \in SO(3)$
   vector $a \in R^3$
   $\boxed{R S(a) R^T = S(Ra) \Rightarrow R S(a) = S(Ra) R}$
4. $b^T S(a)\, b = 0$
   $S(a) \in so(n)$
   vector $b \in R^n$

angular velocity

$\dot{R}(t) = S(\omega(t))\, R(t)$

addition of angular velocities

$\omega^0_{0,2} = \omega^0_{0,1} + R^0_1(t)\, \omega^1_{1,2}$

linear velocity $v = \dot{o}^0_1$

$\dot{p}^0 = \dot{R}^0_1 p^1 + \dot{o}^0_1 = S(\omega^0_{0,1}) R^0_1 p^1 + \dot{o}^0_1 = \omega^0_{0,1} \times r + v$

where $r = R^0_1 p^1 \rightarrow$ vector expressing p^1 in $o_0 x_0 y_0 z_0$

geometrical jacobian

$$T_n^0(q) = \begin{bmatrix} R_n^0(q) & o_n^0(q) \\ 0^T & 1 \end{bmatrix}$$

q, $R_n^0(q)$ & $o_n^0(q)$ are time dependent functions when robot is in motion

$v_n^0 = \dot{o}_n^0 \rightarrow$ linear velocity ; $S(\omega_n^0) = \dot{R}_n^0 \, (R_n^0)^T \rightarrow$ angular velocity

Jacobian matrix: relates joint velocities $\dot{q}$ to end effector velocities v_n^0 & ω_n^0.

$$J(q) = \begin{bmatrix} J_v(q) \\ J_\omega(q) \end{bmatrix} ; \quad \left.\begin{aligned} v_n^0 &= J_v(q)\dot{q} \\ \boxed{\omega_n^0 = J_\omega(q)\dot{q}} \end{aligned}\right\} \xi = J(q)\dot{q}$$

$q \in R^n$

For angular velocity

$$\omega_n^0 = \sum_{i=1}^{n} \rho_i \dot{q}_i z_{i-1}^0$$

$\rho_i = 1$ for revolute joints

$\rho_i = 0$ for prismatic joints

$$\boxed{J_\omega(q) = \begin{bmatrix} \rho_1 z_0^0 & \cdots & \rho_n z_{n-1}^0 \end{bmatrix}}$$

→ note: it only goes until $n-1$

→ when solving such Q. also look if any ROT matrix is given & take proj from it.

For linear velocity

$$\dot{o}_n^0 = \sum_{i=1}^{n} \frac{\partial o_n^0}{\partial q_i} \dot{q}_i \quad ; \quad \boxed{J_v(q) = \begin{bmatrix} \frac{\partial o_n^0}{\partial q_1} & \cdots & \frac{\partial o_n^0}{\partial q_n} \end{bmatrix}}$$

OR

$$\frac{\partial o_n}{\partial q_i} = z_{i-1} \times (o_n - o_{i-1}) \rightarrow \text{revolute joints}$$

$$\frac{\partial o_n}{\partial q_i} = z_{i-1} \rightarrow \text{prismatic joints}$$

Velocity transformations

to change ref. system of jacobian $\rightarrow$ $$J^1(q) = \begin{bmatrix} R_0^1 & 0 \\ 0 & R_0^1 \end{bmatrix} J^0(q)$$

$J^0(q) \rightarrow$ expressed in base cos $o_0 x_0 y_0 z_0$

$J^1(q) \rightarrow$ jacobian expressed in terms of cos $o_1 x_1 y_1 z_1$

velocity of point P_2 rigidly connected to P_1 whose velocity is known.

in fixed cos $\rightarrow$ $$\begin{bmatrix} v_2 \\ \omega_2 \end{bmatrix} = \begin{bmatrix} I & -S(P_{12}) \\ 0 & I \end{bmatrix} \begin{bmatrix} v_1 \\ \omega_1 \end{bmatrix}$$

$P_{12} = P_2 - P_1 \rightarrow$ vector from P_1 to P_2.

in local cos $\rightarrow$ $$\begin{bmatrix} v_2^2 \\ \omega_2^2 \end{bmatrix} = \begin{bmatrix} R_1^2 & -R_1^2 S(P_{12}^1) \\ 0 & R_1^2 \end{bmatrix} \begin{bmatrix} v_1^1 \\ \omega_1^1 \end{bmatrix}$$

Singularities: A config $q \in \mathbb{R}^n$ is called singular if matrix doesn't have full rank.

Ex: $\text{rank}(J(q)) < 6$

or in general $\text{rank}(\underbrace{J}_{n\text{-rows}}(q)) < n \rightarrow$ reduced jacobian matrix.

Determine singular configurations from eqn:

$$\boxed{\det(J(q)) = 0}$$

get singular config q_s & put back into $J(q)$ for examining rank

for robot with 6-degrees of freedom

$$J = \begin{bmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{bmatrix}$$

$J_{11}, J_{12}, J_{21}, J_{22} \rightarrow$ 3×3 ~~rect~~ blocks

$J_{11}, J_{12} \rightarrow J_v$ of body & wrist respectively.

$J_{21}, J_{22} \rightarrow J_\omega$

Analytical jacobian:

$$X = \begin{bmatrix} p(q) \\ \alpha(q) \end{bmatrix} \begin{matrix} \rightarrow \text{pose} \\ \rightarrow \text{orientation} \end{matrix} \qquad \dot{X} = \begin{bmatrix} \dot{p}(q) \\ \dot{\alpha}(q) \end{bmatrix} = J_a(q)\,\dot{q}$$

$$\omega = B(\alpha)\dot{\alpha}$$

$$J_a(q) = \begin{bmatrix} I & 0 \\ 0 & B^{-1}(\alpha) \end{bmatrix} J(q) \quad ; \text{if } B(\alpha) \text{ is invertible}$$

if only boundary condition for position & velocity given → use cubic polynomial

(4)

* **Cubic polynomials**: $q(t) = \sum_{i=0}^{3} a_i t^i = a_0 + a_1 t + a_2 t^2 + a_3 t^3$

$$\dot{q}(t) = a_1 + 2a_2 t + 3a_3 t^2$$

initial conditions: $q(t_0) = q_0 \qquad q(t_f) = q_f$

$\dot{q}(t_0) = V_0 \qquad \dot{q}(t_f) = V_f$

initial cond. will be given, form linear syst. of eqn. & solve for coeff. [gaussian elemination]

boundary cond. for pos, vel & acc → use quintic polynomial

* **quintic polynomials**:

$$q(t) = \sum_{i=0}^{5} a_i t^i = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5$$

$$\dot{q}(t) = a_1 + 2a_2 t + 3a_3 t^2 + 4a_4 t^3 + 5a_5 t^4$$

$$\ddot{q}(t) = 2a_2 + 6a_3 t + 12 a_4 t^2 + 20 a_5 t^3$$

order of polynomial = n.o of boundary conditions -1

Trapezoidal trajectories: - to check satisfaction of constraints easily
- const. acceleration at start & end + const. velocity in between

Parametrization by durations [total duration & duration of acceleration given]

(i) $$q(t) = \begin{cases} q_0 + \frac{1}{2} a (t - t_0)^2 & ; t \in [t_0, t_0 + t_a] \\ q_0 + \frac{1}{2} a t_a^2 + v(t - t_a - t_0) & ; t \in (t_0 + t_a, t_f - t_a) \\ q_f - \frac{1}{2} a (t_f - t)^2 & ; t \in [t_f - t_a, t_f] \end{cases}$$

(ii) $$\dot{q}(t) = \begin{cases} a(t - t_0) & t \in [t_0, t_0 + t_a] \\ V & t \in (t_0 + t_a, t_f - t_a) \\ a(t_f - t) & t \in [t_f - t_a, t_f] \end{cases}$$

(iii) $$\ddot{q}(t) = \begin{cases} a & ; t \in [t_0, t_0 + t_a] \\ 0 & ; t \in (t_0 + t_a, t_f - t_a) \\ -a & ; t \in [t_f - t_a, t_f] \end{cases}$$

(I) with

$V = a t_a$

$$V = \frac{q_f - q_0}{t_f - t_0 - t_a}$$

Parametrization by velocity & Acceleration [Acceleration & velocity given minimal duration $t_f - t_0$ searched]

$\text{sgn}(v) = \text{sgn}(a) = \text{sgn}(q_f - q_0)$

i.e. V & a are +ve for $q_f > q_0$

-ve for $q_f < q_0$

limitations: $- \; t_a \leq V/a$

$- \; \frac{1}{2} a t_a^2 \leq \frac{1}{2} (q_f - q_0)$; i.e. at most $\frac{1}{2}$ dist travelled within t_a.

$$t_a = \min\left\{ \frac{V}{a}, \sqrt{\frac{q_f - q_0}{a}} \right\} \quad ; \quad t_f = t_0 + t_a + \frac{q_f - q_0}{a \, t_a}$$

Use i, ii & iii for computing trajectory.

Synchronization with multiple joints

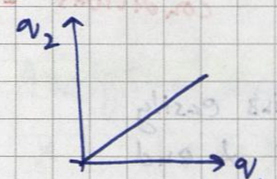
can plan separate trajectory for each joint $q_i(t)$; $i = 1, \dots n$

Without synchronisation:
choose trajectory with min duration $t_{f,i} - t_0$ for each joint.

Time synchronisation: requires all trajectories have same duration.

i) calc. $t_{f,i}$ for each joint simultaneously
ii) choose $t_f = \max_i t_{f,i}$ as duration for all joints.

Phase synchronisation: requires motion from q_0 to q_f to be linear in joint space.



- individual phases with const. acceleration and const. velocity must be synchronised in time.

$$q(t) = (1 - s(t))\, q_0 + s(t)\, q_f$$

$$s(t) : [t_0, t_f] \to [0, 1]$$

$$\dot{q}(t) = \dot{s}(t)(q_f - q_0) \quad ; \quad \ddot{q}(t) = \ddot{s}(t)(q_f - q_0)$$

$$\Rightarrow \dot{s}(t) \le v_s = \min_i \frac{v_i}{|q_{f,i} - q_{0,i}|} \qquad \ddot{s}(t) \le a_s = \min_i \frac{a_i}{|q_{f,i} - q_{0,i}|}$$

plan time optimal traj for $s(t)$ with following boundary conditions.

$s(t_0) = 0 \qquad s(t_f) = 1$

$\dot{s}(t_0) = \dot{s}(t_f) = 0$ & v_s, a_s limits for vel. & acc. respectively.

For phase sync. use (I) from page (4) to calc. new v & a with synchronised times t_a & t_f of the 2 joints. After calc. substitute into (i) (ii) & (iii) for trajectories which are phase sync.

Time sync: Can also time scale the slower trajectory's time scale & trajectory itself using ratio b/w t_{f_1} & t_{f_2} accordingly

$\tilde{t}$ = desired time-line Ex: $[0, 4.5]$ $\to t_{f_1}$; $\tilde{t} = [0, 4.5]$

$$t(\tilde{t}) = \tilde{t}\, \frac{t_{f,2}}{t_{f,1}} = \lambda \tilde{t} \quad \Rightarrow \quad \tilde{t}_a = \frac{t_a}{\lambda} \; ; \; \tilde{t}_f = \frac{t_f}{\lambda}$$

&

$$\tilde{a} = a \lambda^2 \qquad \tilde{v} = v \lambda$$

Trajectories with intermediate points

$q(t_i) = p_i \; ; \; i = 0 \dots N$ (5)

- Interpolation with cubic polynomials

$$q_i(t) = \frac{a_j}{6\Delta t_i}(t-t_i)^3 + \left(\frac{p_j}{\Delta t_i} - \frac{a_j \Delta t_i}{6}\right)(t-t_i) + \frac{a_i}{6\Delta t_i}(t_j-t)^3 + \left(\frac{p_i}{\Delta t_i} - \frac{a_i \Delta t_i}{6}\right)(t_j - t)$$

$j = i+1$

$\Delta t_i = t_j - t_i$

$$\begin{bmatrix} \frac{\Delta t_0}{3} & \frac{\Delta t_0}{6} & 0 & \cdots & 0 \\ \frac{\Delta t_0}{6} & \frac{\Delta t_0 + \Delta t_1}{3} & \frac{\Delta t_1}{6} & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & \frac{\Delta t_{N-2}}{6} & \frac{\Delta t_{N-2}+\Delta t_{N-1}}{3} & \frac{\Delta t_{N-1}}{6} \\ 0 & \cdots & 0 & \frac{\Delta t_{N-1}}{6} & \frac{\Delta t_{N-1}}{3} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{N-1} \\ a_N \end{bmatrix} = \begin{bmatrix} \frac{p_1 - p_0}{\Delta t_0} - v_0 \\ \frac{p_2 - p_1}{\Delta t_1} - \frac{p_1 - p_0}{\Delta t_0} \\ \vdots \\ \frac{p_N - p_{N-1}}{\Delta t_{N-1}} - \frac{p_{N-1} - p_{N-2}}{\Delta t_{N-2}} \\ v_N - \frac{p_N - p_{N-1}}{\Delta t_{N-1}} \end{bmatrix}$$

Thomas Algorithm → express a_0 in terms of a_1
a_1 '' of a_2 ... compute last row a_N & seq. insert

- Linear segments with parabolic blends

1^{st} interval : p_0 to p_1

$$v_0 = \frac{p_1 - p_0}{\Delta t_0 - \frac{1}{2} t_{a,0}} = \boxed{a_0 t_{a,0}}$$

$$t_{a,0} = \Delta t_0 - \sqrt{\Delta t_0^2 - 2\frac{p_1 - p_0}{a_0}}$$

$a_0 = sgn(p_1 - p_0) \cdot a$
or
$a_0 = \frac{v_0}{t_a}$

last interval : p_{N-1} to p_N

$$v_{N-1} = \frac{p_N - p_{N-1}}{\Delta t_{N-1} - \frac{1}{2} t_{a,N}} = \boxed{-a_N t_{a,N}}$$

$$t_{a,N} = \Delta t_{N-1} - \sqrt{\Delta t_{N-1}^2 + 2\frac{p_N - p_{N-1}}{a_N}}$$

$a_N = -sgn(p_N - p_{N-1}) \cdot a$

inner intervals

$$v_i = \frac{p_j - p_i}{\Delta t_i}$$

$$t_{a,j} = \frac{|v_j - v_i|}{2a}$$

or

$$a_j = \frac{v_j - v_i}{2 t_a}$$ → if $\frac{1}{2}$ blend duration is mentioned

$\Delta t_i = t_j - t_i \quad j = 1, \dots, N-1$

P.T.O. for position profile

$$q_i(t) = p_i + v_i(t-t_i) + \begin{cases} \frac{1}{2} a_i (t-t_i-t_{a,i})^2 & \text{if } t \in [t_i, t_i + t_{a,i}] \\ 0 & \text{if } t \in (t_i + t_{a,i}, t_j - t_{a,j}) \\ \frac{1}{2} a_j (t-t_j+t_{a,j})^2 & \text{if } t \in [t_j - t_{a,j}, t_j] \end{cases}$$

Trajectories in task space

Position: $p(t) = (1-s(t))\, p_0 + s(t)\, p_f$

$p_0 \rightarrow$ start position $\qquad s(t) \rightarrow$ function

$p_f \rightarrow$ final position $\qquad s(t_0)=0$ & $s(t_f)=1$

Orientation: $\alpha(t) = (1-s(t))\,\alpha_0 + s(t)\,\alpha_f$

euler angles $\rightarrow$
$\alpha_0 = [\phi_0, \theta_0, \psi_0]^T$ corresponding to R_0
$\alpha_f = [\phi_f, \theta_f, \psi_f]^T$ " " R_f

$R_f = R_0 R \longrightarrow R = R_0^T R_f$

$\hookrightarrow$ parametrized by angle axis $\Rightarrow R = e^{S(k)\theta}$

plan time parametrisation $s(t)$. $s(t_0)=0$; $s(t_f)=1$

$$R(t) = R_0\, e^{S(k)\theta s(t)} = R_0\, R^{s(t)}$$

using quaternions

Q_0 corresponding to R_0

Q_f " " R_f

$Q = Q_0^* Q_f \rightarrow$ describes relative orientation R

interpolated quaternion $\boxed{Q(t) = Q_0 Q^{s(t)} = Q_0 (Q_0^* Q_f)^{s(t)}}$

$$\boxed{Q^s = \left(\cos \frac{\theta s}{2}, \sin \frac{\theta s}{2} k\right)}$$

Conj. of Quatenion

if $Q = (q_0, q_1, q_2, q_3)$

$Q^* = (q_0, -q_1, -q_2, -q_3)$

Linear Control

$\tau_m = k_m i_a \rightarrow$ torque due to armature current i_a through the rotor.

1st principle: force act on a current carrying conductor in a magnetic field.

2nd principle: motion of a conductor in magnetic field causes voltage. proportional to velocity, ω_m.

$u_v = k_v \omega_m$

$\underbrace{k_m}_{\text{motor torque const.}} = \underbrace{k_v}_{\text{motor velocity const.}}$; if SI units used

Model of Permanent magnet DC motor with transmission

voltage balance $\rightarrow L_a \frac{di_a}{dt} + R_a i_a = u_a - u_v = u_a - k_v \omega_m$; $T_a = \frac{L_a}{R_a}$

torque balance $\rightarrow J_m \frac{d\omega_m}{dt} + b_m \omega_m = \tau_m - \frac{1}{r_m}\tau_l = k_m i_a - \frac{1}{r_m}\tau_l$ → load torque

; $T_m = \frac{J_m}{b_m}$

Reduced model $\rightarrow J_m \frac{d\omega_m}{dt} + \left(b_m + \frac{k_m k_v}{R_a}\right)\omega_m = \frac{k_m}{R_a} u_a - \frac{1}{r_m}\tau_l$

Linear 2nd order diff. eqn.

$\cancel{J_m \theta}$ $J_m \ddot{\theta}_m + b\dot{\theta}_m = u - d$; $b = b_m + \frac{k_m k_v}{R_a}$, $u = \frac{k_m}{R_a} u_a$,

$d = \frac{\tau_l}{r_m}$ → load torque considered by disturbance.

Rigid body dynamics: dynamical behaviour of manipulator.

$$\underbrace{D(q)}_{\text{inertia matrix}}\ddot{q} + \underbrace{C(q,\dot{q})}_{\text{corriolis \& centrifugal forces}}\dot{q} + \underbrace{g(q)}_{\text{gravity vector}} = \underbrace{\tau}_{\text{load torque for motor}}$$

↳ coupling effect due to n.o. of joints.

Actuator dynamics:

$$\boxed{J_m \ddot{\theta}_m + B\dot{\theta} = u - R_m^{-1}\tau}$$

↳ load torque is interface to rigid body dynamics

$J_m = \text{diag}(J_{m,1} \dots J_{m,n})$

$B = \text{diag}(b_1 \dots b_n)$

$R_m = \text{diag}(r_{m,1} \dots r_{m,n})$ → gear ratios ; $\tau = [\tau_1 \dots \tau_n]^T$

incase of rigid coupling: $\underbrace{\theta_m}_{\text{motor position}} = R_m \underbrace{q}_{\text{joint position}} \Rightarrow q = R_m^{-1}\theta_m$

Combination of Rigid body and Actuator dynamics:

in terms of joint pos:

$$\left(D(q) + R_m^2 J_m\right)\ddot{q} + \left(C(q,\dot{q}) + R_m^2 B\right)\dot{q} + g(q) = R_m u$$

interms of motor pos:

$$\left(R_m^{-1} D(q) R_m^{-1} + J_m\right)\ddot{\theta}_m + \left(R_m^{-1} C(q,\dot{q}) R_m^{-1} + B\right)\dot{\theta}_m + R_m^{-1} g(q) = u$$

Split of inertia matrix for decoupled model: $D(q) = \bar{D} + \Delta D(q)$

$$\boxed{\bar{D} = diag(d_1 \dots d_n) \rightarrow \text{const. part}}$$

$\Delta D(q) = D(q) - \bar{D} \rightarrow$ non-linear config-dependent part

Combine RBD dynamics & actuator dyn. to get decoup. model

decoupled model: Trick: separate out linear terms on LHS of RBD

$$\boxed{M\ddot{\theta}_m + B\dot{\theta}_m = u - d(q,\dot{q},\ddot{q})} \quad - *$$

$$\boxed{M = R_m^{-2}\bar{D} + J_m}$$

$$d(q,\dot{q},\ddot{q}) = R_m^{-1}\left(\Delta D(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q)\right) \rightarrow \text{non-linear part}$$

Influence of non-linear part reduced by high gear ratios.

Conditions for nonlinear robot model - *

1. gear ratios R_m are significantly larger than 1
2. Actuators exhibit good rejection of disturbances
3. vel. & accelerations & thus required torques are not too high.

Linear 2^{nd} order systems

Ex: Mass- spring damper system

dynamics: $m\ddot{x}(t) + b\dot{x}(t) + kx(t) = f(t)$

parameters: $m \rightarrow$ mass
$b \rightarrow$ friction related damping
$k \rightarrow$ spring stiffness

$ms^2 + bs + k = 0$ characteristic eqn.

zeros: $s_{1,2} = \frac{-b}{2m} \pm \frac{\sqrt{b^2 - 4mk}}{2m}$

System behaviour in different cases

1. $b^2 - 4mk > 0$; friction dominates ; 2 distinct real zeros

$$x(t) = c_1 e^{s_1 t} + c_2 e^{s_2 t}$$

2. $b^2 - 4mk < 0$; spring dominates ; 2 complex conjugate zeros

Under damped

$s_{1,2} = \tau \pm \mu i$; $\tau = \frac{-b}{2m}$; $\mu = \frac{\sqrt{4mk - b^2}}{2m}$

solutions i) use sol. from case 1 i.e. $x(t) = c_1 e^{s_1 t} + c_2 e^{s_2 t}$

ii) $x(t) = c_1 e^{\lambda t} \cos(\mu t) + c_2 e^{\lambda t} \sin(\mu t)$; from $e^{jx} = \cos x + j \sin x$

iii) If parameters are defined as

$$c_1 = r\cos\delta \qquad c_2 = r\sin\delta$$

with polar coordinates

$$r = \sqrt{c_1^2 + c_2^2} \qquad \delta = \mathrm{atan2}(c_1, c_2)$$

then $$x(t) = r e^{\lambda t} \cos(\mu t - \delta)$$

3. $b^2 = 4mk$; friction & stiffness balanced ; double zero [Critical damping]

$$s_{1,2} = -\frac{b}{2m}$$

$$x(t) = (c_1 + c_2 t) e^{s_{1,2} t}$$

4. $b = 0$; zeros are purely imaginary [undamped]

$$s_{1,2} = \pm \mu i \quad ; \quad \mu = \sqrt{\frac{k}{m}}$$

$$x(t) = c_1 \cos(\mu t) + c_2 \sin(\mu t)$$

system is unstable if one of the 2 zeros $s_{1,2}$ had a real +ve part.

Alternative characteristic equation

$$s^2 + 2\xi\omega_n s + \omega_n^2 = 0$$

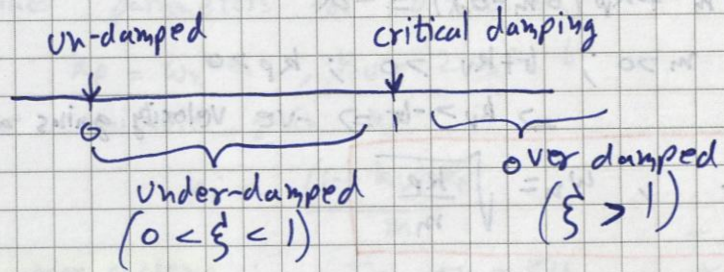
$\boxed{\xi = \frac{b}{2\sqrt{km}} \rightarrow \text{damping factor}}$ $\boxed{\omega_n = \sqrt{\frac{k}{m}} \rightarrow \text{natural frequency}}$

$$s_{1,2} = -\omega_n\left(\xi \pm \sqrt{\xi^2 - 1}\right)$$

Can tell damping behaviour just by looking at ξ.

For sys. to converge to 0 i.e. $\lim_{t\to\infty} x(t) = 0$

then $\underbrace{m, b, k}_{\text{have to be greater than zero}} > 0$



Constants c_1 & c_2 found from initial conditions $x(0)$ and $\dot{x}(0)$

If values of ξ & ω_n are not known/known in terms of some variable, rec to find $x(t)$

use $x(t) = c_1 e^{-s_1 t} + c_2 e^{-s_2 t}$

$$= c_1 e^{-\omega_n(\xi + \sqrt{\xi^2-1})t} + c_2 e^{-\omega_n(\xi - \sqrt{\xi^2-1})t}$$

Control design

model for 1 joint: $m\ddot{\theta}_m + b\dot{\theta}_m = u - d$ → linear decoupled system

motor pos. θ_m can be converted to joint pos using

$$\dot{\theta}_m = r_m \dot{q} \quad ; \quad \ddot{\theta}_m = r_m \ddot{q}$$

1st control aim: stabilization of joints in set-point θ_d.

i.e. $\theta_m = \theta_d$; $\dot{\theta}_m = 0$; $\ddot{\theta}_m = 0$. — aim

P-control: user proportional part for position error $\theta_m - \theta_d$

$$u = -k_p(\theta_m - \theta_d) \rightarrow \text{control law}$$

$$m\ddot{\theta}_m + b\dot{\theta}_m + k_p(\theta_m - \theta_d) = -d \rightarrow \text{dynamics}$$

for stability $m > 0$, $b > 0$ & $k_p > 0$

if $d = 0$ control aim is achieved

if $d \neq 0$ & $d = \text{const.}$ then $\lim_{t\to\infty} \dot{\theta}_m(t) = 0$ & $\lim_{t\to\infty} \ddot{\theta}_m(t) = 0$

steady state deviation $\boxed{\lim_{t\to\infty} \theta_m(t) - \theta_d = \frac{-d}{k_p}}$

For steady state deviation ↓ → k_p↑

but

$$\xi = \frac{b}{2\sqrt{k_p m}} \Rightarrow \xi < 1 \text{ [overshooting]}$$

PD-control: state feedback controller

$$\frac{d}{dt}(\theta_m(t) - \theta_d) = \dot{\theta}_m \rightarrow \text{time derivative of control error = velocity.}$$

$$u = -k_p(\theta_m - \theta_d) - k_v\dot{\theta}_m \rightarrow \text{control law}$$

$$m\ddot{\theta} + (b + k_v)\dot{\theta}_m + k_p(\theta_m - \theta_d) = -d$$

for stability $m > 0$; $b + k_v > 0$; $k_p > 0$

$\Rightarrow k_v > -b \Rightarrow$ -ve velocity gains admissible

$$\boxed{\xi = \frac{b + k_v}{2\sqrt{k_p m}} \quad , \quad \omega_n = \sqrt{\frac{k_p}{m}}}$$

$\xi = 1$ for fastest response ; ω_n can be used as single design parameter

$$k_p = \omega_n^2 m \quad ; \quad k_v = 2\omega_n m - b$$

PID-control

$$u = -k_p(\theta_m - \theta_d) - k_v \dot{\theta}_m - k_i \int_0^t (\theta_m - \theta_d)\, d\tau$$

$$m\ddot{\theta}_m + (b + k_v)\dot{\theta}_m + k_p(\theta_m - \theta_d) + k_i \int_0^t (\theta_m - \theta_d)\, d\tau = -d$$

→ differentiate eqn. to examin closed loop behavior

$\because d = \text{const} \quad \dot{d} = 0$

$$m\,\theta_m^{(3)} + (b + k_v)\ddot{\theta}_m + k_p \dot{\theta}_m + k_i(\theta_m - \theta_d) = 0$$

$\lim_{t\to\infty} k_i(\theta_m(t) - \theta_d) = 0 \Rightarrow \lim_{t\to\infty} \theta_m(t) = \theta_d$ → steady state condition

For stability, check: [~~routh~~ Hurwiz criterion]

$$\boxed{\begin{array}{l} m > 0 \\ b + k_v > 0 \\ (b + k_v)k_p - m k_i > 0 \end{array}}$$ → conditions have to hold for stability

$$\Rightarrow \boxed{k_i < \frac{(b + k_v)\,k_p}{m}}$$

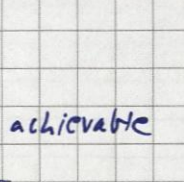
Design Guideline

- eliminate stationary deviation with integral part
- k_i should be small to not influence transient behaviour
- in theory large values can be used for fast behaviour. In practice achievable vel. is limited, due to max torque/max current.
- neglected elasticity of joints & links. resonant freq $= \omega_r = \sqrt{k_r/J_r}$
  To avoid undesired excitations natural freq. should not be higher than half of lowest resonance frequency.

1. determine lowest ω_r & choose ω_n
   $$\omega_n = \frac{1}{2}\omega_r \quad \text{with } \xi = 1 \text{ for fastest resp. w/o overshooting.}$$
2. determine parameters for PD-control
   $$k_p = \omega_n^2 m \quad , \quad k_v = 2\omega_n m - b$$
3. choose suitable gain k_i
   $$k_i < \frac{(b + k_v)\,k_p}{m}$$

Linear 1st order systems : $T\dot{y} + y = Ku$

T → time const.
K → gain
y → o/p ; u → i/p



if u(t) is step input at t=0 then,

$$y(t) = K(1 - e^{-t/T}) \ , \ t \geq 0 \quad ; \quad y(T) = 0.63K$$

No oscillations

desired state $y_d \Rightarrow T_d \dot{y} + y - y_d = 0$

$$u = -\frac{T}{K T_d}(y - y_d) - \frac{1}{K T_d}\int_0^t (y - y_d)\, d\tau \longrightarrow \text{PI controller}$$

$$\underbrace{T(T_d\dot{y} + y - y_d)}_{\dot{z}} + \underbrace{T_d y + \int_0^t (y - y_d)\, d\tau}_{z} = 0 \Longrightarrow T\dot{z} + z = 0$$

→ sol. converges to $z = 0$ if $T > 0$

$\Rightarrow y$ converges to y_d

current control [cascaded]

$$L_a \frac{di_a}{dt} + R_a i_a = u_a - k_b \omega_m \quad \rightarrow 1^{st} \text{ order lag element}$$

$$T = \frac{L_a}{R_a} \qquad \text{gain factor} = \frac{1}{R_a}$$

→ compensate motor vel

$$u_a = k_b \omega_m - \underbrace{\frac{L_a}{T_i}(i_a - i_{a,d})}_{P} - \underbrace{\frac{R_a}{T_i}\int_0^t (i_a - i_{a,d})\,d\tau}_{I} \quad ; \quad T_i \rightarrow \text{desired time const.}$$

$$T_i \frac{di_a}{dt} + i_a - i_{ad} = 0 \quad \rightarrow \text{closed loop dynamics.}$$

Choose T_i so that it is faster than mechanical subsystem.

$$J_m \frac{d\omega_m}{dt} + b_m \omega_m = k_m i_a - \frac{1}{r_m}\tau_l$$

assuming $i_a \approx i_{a,d}$

$$u = k_m i_{ad} \quad ; \quad T = \frac{J_m}{b_m}$$

velocity control [cascaded]

$$m\ddot{\theta}_m + b\dot{\theta}_m = u - d \quad \text{is } 1^{st} \text{ order lag element}$$

→ of linear 2^{nd} order decoupled sys.

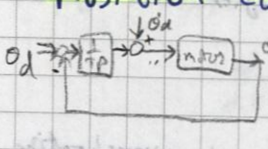
with, $\omega_n = \dot{\theta}_m$ with $T = \frac{m}{b}$ & gain factor $= \frac{1}{b}$

$$\boxed{u = -\frac{m}{T_v}(\dot{\theta}_m - v) - \frac{b}{T_v}\int_0^t (\dot{\theta}_m - v)\,d\tau} \quad ; \quad T_v \rightarrow \text{desired time const}$$

$v \rightarrow$ setpoint

without d, closed loop dynamics $\Rightarrow T_v \ddot{\theta}_m + \dot{\theta}_m - v = 0$

Position control [cascaded]



$$\boxed{v = \underbrace{\dot{\theta}_d}_{\text{desired vel}} - \underbrace{\frac{1}{T_p}(\theta_m - \theta_d)}_{P}} \quad ; \quad T_p \rightarrow \text{desired Time const}$$

$\theta_d \rightarrow$ desired position

$\dot{\theta}_d \rightarrow$ " velocity

$$T_v \ddot{\theta}_m + (\dot{\theta}_m - \dot{\theta}_d) + \frac{1}{T_p}(\theta_m - \theta_d) = 0$$

$$\omega_n = \sqrt{\frac{1}{T_p T_v}} \qquad \xi = \sqrt{\frac{T_p}{4T_v}}$$

critical damping $\Rightarrow T_p = 4T_v$

$$\boxed{T_p = \frac{2\xi}{\omega_n}, \quad T_v = \frac{1}{2\xi\omega_n}}$$

Advantage over PID controller is both critical damping & no steady state error is achievable.

- closed loop is still 2^{nd} order system.
- desired velocity v & desired torque u appear as explicit o/p of pos controller. ⇒ implementation of limits using saturation functions possible.
- antiwindup strategy required for integrator.

Relation to non-cascaded control

$$u = \frac{b}{T_v} c - \underbrace{\overbrace{\left(\frac{m}{T_v T_p} + \frac{b}{T_v}\right)}^{P}}_{K_p} (\theta_m - \theta_d) - \underbrace{\overbrace{\frac{m}{T_v}}^{d}}_{K_d} (\dot\theta_m - \dot\theta_d) - \underbrace{\overbrace{\frac{b}{T_v T_p}}^{I}}_{K_i} \int_0^t (\theta_m - \theta_d)\, d\tau$$

$$\boxed{\begin{aligned} K_p &= \frac{m}{T_v T_p} + \frac{b}{T_v} = \omega_n^2 m + 2\xi\omega_n b \\ K_v &= \frac{m}{T_v} = 2\xi\omega_n m \\ k_i &= \frac{b}{T_v T_p} = \omega_n^2 b \end{aligned}}$$

Task space control

Definition of control error

(6x1 vector) $e = \begin{bmatrix} e_p \\ e_o \end{bmatrix}$ 3x1 vector, 3x1 vector

$e_p = p(t) - p_d(t)$ → Position error

$e_o(\alpha, \alpha_d) = \alpha - \alpha_d$ → orientation error

Alternatively for e_o

$R_\Delta = R R_d^T$; R_Δ → relative rotation matrix [rot. matrix error that brings current rot. to desired rotation]

$\Rightarrow R = R_\Delta R_d$

→ parametrise with axis k_Δ with $\|k\| = 1$ & angle θ_Δ

then $e_o(R, R_d) = \frac{1}{2}\left(r_{d,1} \times r_1 + r_{d,2} \times r_2 + r_{d,3} \times r_3\right)$

with Quaternions for e_o

$q = (q_0, q)$ $q_d = (q_{0,d}, q_d)$

$$e_o(q, q_d) = \sin\left(\frac{\theta_\Delta}{2}\right) k_\Delta = q_{d,0}\, q - q_0\, q_d + q_d \times q$$

Resolved Acceleration control

$\ddot e + K_{x,v}\dot e + K_{x,p} e \overset{!}{=} 0$ → closed loop dynamics

$K_{x,p} = \omega_n^2 I$; $K_{x,v} = 2\xi\omega_n I$ → diagonal matrices } asymptotically converges to 0

$e = x - x_d$ with $x = [p^T, \alpha^T]^T$; $x_d = [p_d^T, \alpha_d^T]^T$

$\ddot x \overset{!}{=} \ddot x_d - K_{x,v}\dot e - K_{x,p} e$ & $\ddot x = J_a(q)\ddot q + \dot J_a(q)\dot q$

$$\Rightarrow \ddot q = J_a^{-1}(q)\left(\ddot x - \dot J_a(q)\dot q\right) = J_a^{-1}(q)\left(\ddot x_d - K_{x,v}\dot e - K_{x,p} e - \dot J_a(q)\dot q\right)$$

↳ necessary joint accelerations

with linear decoupled law $M\ddot\theta_m + B\dot\theta_m = u - d$ & $\dot\theta_m = R_m \dot q$

control law is obtained as

$$u = M R_m J_a^{-1}(q)\left(\ddot x_d - K_{x,v}\dot e - K_{x,p} e - \dot J_a(q)\dot q\right) + B\dot\theta_m$$

pg 103 for const disturbance eqns [eqn 4.102]

s.s. deviation for $d = $ const

$$\lim_{t\to\infty} e = -K_{x,p}^{-1} J_a(q) R_m^{-1} M^{-1} d$$

Resolved velocity control

$$T_V \ddot{\theta}_m + \dot{\theta}_m - V = 0 \rightarrow \text{closed loop dynamics}$$

$V \rightarrow$ velocity set-points

$$\dot{\xi}_d = [V_d^T, w_d^T] \rightarrow \text{desired task space velocity}$$

$$V = \underset{\substack{\uparrow \\ \text{gear ratio}}}{R_m} J^{-1}(q)\, \xi_d \rightarrow \text{necessary joint velocity}$$

p-control as outer loop.

$$\xi_d = -K_{x,p}\, e$$

Transposed Jacobian control

Doesnt require inversion of Jacobian

Transposed jacobian transforms forces & torques at end effector to joint torques.

$$u = -R_m J(q)^T K_{x,p}\, e - k_V \dot{\theta}_m$$

with damping

$$u = -R_m J(q)^T (K_{x,p} e + K_{x,v} \xi)$$

If orientation error is computed using euler angles then use analytical jacobian.

$$u = -R_m J_a(q)^T (K_{x,p} e + K_{x,v} \dot{e})$$

Force Control

Direct force control.

$$m\ddot{x} + b\dot{x} = u - f_e - d \rightarrow \text{dynamical model}$$

Position control & contact force

$$u = -k_p (x - x_d) - k_V \dot{X}$$

$$\Rightarrow m\ddot{x} + (b + k_V)\dot{x} + k_p(x - x_d) = -f_e$$

$$f_e = k_e (x - x_e) \quad ; \quad x_e \rightarrow \text{ref position of environment}$$

$$\Rightarrow m\ddot{x} + (b + k_V)\dot{x} + k_p(x - x_d) + k_e(x - x_e) = 0$$

for $t \rightarrow \infty$ with $\dot{x} = \ddot{x} = 0$

equilibrium pos $\rightarrow$ $$x = \frac{k_p}{k_p + k_e} x_d + \frac{k_e}{k_p + k_e} x_e$$

for stiff env. $k_e \gg k_p \Rightarrow x \approx x_e$

for compliant env $k_p \gg k_e \Rightarrow x \approx x_d$

PD force control

aim: to stabilize contact force f_e at desired setpoint f_d.
force control achieved by position control with desired position.

$$f_d = k_e(x_d - x_e) \Leftarrow x_d = x_e + \frac{f_d}{k_e}$$

deviation of measured force f_e vs desired force

$$f_e - f_d = k_e(x - x_e) - k_e(x_d - x_e) = k_e(x - x_d)$$

since x_e & f_d are const.

$$\frac{d}{dt}(f_e - f_d) = \dot{f}_e = k_e \dot{x} \qquad \frac{d^2}{dt}(f_e - f_d) = \ddot{f}_e = k_e \ddot{x}$$

$$(x - x_d) = \frac{1}{k_e}(f_e - f_d), \quad \dot{x} = \frac{1}{k_e}\dot{f}_e \quad \ddot{x} = \frac{1}{k_e}\ddot{f}_e$$

$$\boxed{u = f_e - \frac{k_p}{k_e}(f_e - f_d) - k_v \dot{x}}$$

→ augment $-\frac{k_i}{k_e}\int_0^t (f_e - f_d)d\tau$ for zero steady state error

$$m\ddot{x} + (b + k_v)\dot{x} + \frac{k_p}{k_e}(f_e - f_d) = 0$$

f_e in u is feed fwd to get rid of ext. force hence depending on model it could be even $0.5 f_e$.

Steady state error

$$\lim_{t\to\infty} f_e(t) - f_d = -\frac{k_e}{k_p} d$$

Alternative: use feed forward of desired force f_d instead of compensating measured force f_e.

$$u = f_d - \frac{k_p}{k_e}(f_e - f_d) - k_v \dot{x}$$

$$m\ddot{x} + (b + k_v)\dot{x} + \left(\frac{k_p}{k_e} + 1\right)(f_e - f_d) = -d$$

steady state error

$$\lim_{t\to\infty} f_e(t) - f_d = \frac{-k_e}{k_p + k_e} d$$

Indirect force control

Impedance control

- robot reacts to external forces like mass-spring damper system

$$e = x - x_d \ ; \ \dot{e} = \dot{x} - \dot{x}_d \ ; \ \ddot{e} = \ddot{x} - \ddot{x}_d$$

closed loop dynamics → $m_d \ddot{e} + b_d \dot{e} + k_d e \stackrel{!}{=} -f_e$ $[d = 0 \text{ here}]$

$$\Rightarrow \ddot{x} = \ddot{x}_d - \frac{1}{m_d}(b_d \dot{e} + k_d e + f_e)$$

$$u = m\left(\ddot{x}_d - \frac{1}{m_d}(b_d \dot{e} + k_d e + f_e)\right) + b\dot{x} + f_e \rightarrow \text{control law}$$

Admittance control [Position based admittance control]

$$u = m\ddot{x}_c + b\dot{x}_c - k_v(\dot{x} - \dot{x}_c) - k_p(x - x_c)$$

closed loop dynamics $\rightarrow m(\ddot{x} - \ddot{x}_c) + (b + k_v)(\dot{x} - \dot{x}_c) + k_p(x - x_c) = -f_e$

dynamical system $\rightarrow m_d(\ddot{x}_c - \ddot{x}_d) + b_d(\dot{x}_c - \dot{x}_d) + k_d(x_c - x_d) = -f_e$

Hybrid force/motion control

$$\begin{array}{c|c} V & \omega \\ f & \mu \end{array} \quad \text{complementary}$$

For Natural constraints

- if we can move in a direction, no force can be exerted
- if we can rotate in a direction, no torque can be exerted

S_x & S_f → selection matrices [select based on artificial constraints]

S_x → 1 if motion possible
0 if motion not-possible

$S_x = \text{diag}(\underset{\downarrow}{v_x}\ \underset{\downarrow}{v_y}\ \underset{\downarrow}{v_z}\ \underset{\downarrow}{\omega_x}\ \underset{\downarrow}{\omega_y}\ \underset{\downarrow}{\omega_z})$

S_f → choose S_f complementary to S_x. $S_f = \text{diag}(f_x\ f_y\ f_z\ \mu_x\ \mu_y\ \mu_z)$

Artificial constraints = set points / desired values